

Netflix Titles Analysis Report

This report provides a comprehensive, data-driven analysis of the Netflix Titles dataset, evaluating the metrics and trends presented in the Power BI dashboard and cross-referencing them with the source files: [Netflix_analysis.csv](#), [Data_cleaning.ipynb](#), and [SQL.sql](#).

1. Executive Summary

The dataset comprises **9,827 titles** spanning more than a century (from **April 17, 1902** to **July 3, 2024**). Below is a summary of the core baseline metrics shown in the dashboard:

Metric	Dashboard Value	Analysis Summary
Average Popularity	40.33	Right-skewed distribution. Most titles are clustered at lower popularity (<35), with a few massive blockbusters.
Average Rating	6.44 / 10	The rating is slightly dragged down by 100 titles with 0-vote placeholders. Excluding these, the actual average is 6.51 .
Average Vote Count	1.39K (1,392.8)	Driven heavily by popular classics and blockbusters with vote counts exceeding 20,000.

[!NOTE]

The dashboard presents the raw averages. Cleaning the dataset by removing unreleased or unvoted titles shifts the average rating upward and provides a more accurate representation of viewer sentiment.

2. Key Insights & Relationship Analysis

A. Popularity vs. Engagement (Vote Count)

- Correlation ($r = 0.137$):** There is a **very weak linear correlation** between a title's current popularity and its total vote count.
- Insight:** Popularity is a dynamic, transient metric reflecting current search/streaming activity, whereas vote count is a cumulative measure of long-term engagement.
 - Example:* **Inception** (2010) has a moderate popularity score of **127.13** but leads the dataset with **31,077 votes**.

- Conversely, **Spider-Man: No Way Home** (2021) has an outlier popularity score of **5,083.95** (the highest in the dataset) but a relatively modest **8,940 votes** because it is a newer release.

B. Popularity vs. Rating

- **Correlation ($r = 0.054$):** There is **no linear correlation** between popularity and ratings. Highly popular/trending titles do not systematically receive better or worse ratings.

C. Vote Count vs. Rating (Audience Size vs. Reception)

- **Correlation ($r = 0.254$):** A **weak-to-moderate positive correlation** exists.
- **Insight:** Highly voted titles (e.g., >10K votes) almost exclusively sit in the **7.0 to 9.0 rating range**. Titles with very low vote counts show extreme rating variance (ranging from 1.0 to 10.0). This suggests that widely-viewed films that stand the test of time tend to be well-regarded, while poorly-received films rarely accumulate massive voting volumes.

graph TD

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VC[High Vote Count] --> R[Consistent Ratings: 7.0 - 9.0]
LVC[Low Vote Count] --> RV[High Rating Variance: 1.0 - 10.0]
P[High Popularity] -->|No Correlation| RT[Rating Independent]
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3. Top Performers & Outliers

Top 5 Most Popular Titles

1. **Spider-Man: No Way Home** (2021) — Popularity: **5,083.95** | Rating: **8.30** | Votes: **8,940**
2. **The Batman** (2022) — Popularity: **3,827.66** | Rating: **8.10** | Votes: **1,151**
3. **No Exit** (2022) — Popularity: **2,618.09** | Rating: **6.30** | Votes: **122**
4. **Encanto** (2021) — Popularity: **2,402.20** | Rating: **7.70** | Votes: **5,076**
5. **The King's Man** (2021) — Popularity: **1,895.51** | Rating: **7.00** | Votes: **1,793**

Top 5 Most Engaged Titles (Highest Vote Count)

1. **Inception** (2010) — Votes: **31,077** | Rating: **8.40**
2. **Interstellar** (2014) — Votes: **27,975** | Rating: **8.40**
3. **The Dark Knight** (2008) — Votes: **26,994** | Rating: **8.50**
4. **The Avengers** (2012) — Votes: **26,490** | Rating: **7.70**
5. **Deadpool** (2016) — Votes: **26,390** | Rating: **7.60**

Top 5 Critically Acclaimed Titles (Rating ≥ 500 votes)

- 1. **The Godfather** (1972) — Rating: **8.70** | Votes: **15,614**
- 2. **The Shawshank Redemption** (1994) — Rating: **8.70** | Votes: **20,911**
- 3. **Dilwale Dulhania Le Jayenge** (1995) — Rating: **8.70** | Votes: **3,474**
- 4. **Gabriel's Inferno Part III** (2020) — Rating: **8.60** | Votes: **946**
- 5. **Schindler's List** (1993) — Rating: **8.60** | Votes: **12,474**

4. Genre Deep-Dive

Since titles can belong to multiple genres, splitting them reveals the true distribution across **19 unique genres**:

- **Volume Leader: Drama** dominates Netflix's library with **3,744 titles**, followed by **Comedy (3,031)** and **Action (2,686)**.
- **Popularity Leader: Adventure** leads in average popularity at **53.74**, closely followed by **Fantasy (53.08)** and **Animation (52.43)**.
- **Rating Leader: History** has the highest average rating at **6.97**, followed closely by **War (6.95)** and **Music (6.88)**.
- **Underperforming Genre: Horror** has the lowest average rating (**5.94**), despite being highly populated (1,470 titles) and relatively popular (avg. popularity: 38.26).

Title Volume by Genre:

Drama	===== 3,744
Comedy	===== 3,031
Action	===== 2,686
Thriller	===== 2,488
Adventure	===== 1,853
Romance	===== 1,476
Horror	===== 1,470
Animation	===== 1,439
Family	===== 1,414
Fantasy	===== 1,308
Sci-Fi	===== 1,273
Crime	===== 1,242
Mystery	===== 773
History	===== 427
War	===== 308
Music	===== 295
Documentary	===== 215

TV Movie == 214

Western = 137

5. Language Analysis

English (en) is the dominant language in this dataset, accounting for **7,570 titles** (~77%). However, non-English titles show strong metrics in both average rating and popularity:

Top Languages by Average Popularity (Min. 10 titles)

1. **Hindi** (hi) – Avg. Popularity: **85.93** (26 titles)| Avg. Rating: **6.63**
2. **Turkish** (tr) – Avg. Popularity: **71.63** (15 titles)| Avg. Rating: **6.45**
3. **Thai** (th) – Avg. Popularity: **60.87** (17 titles)| Avg. Rating: **6.54**
4. **Polish** (pl) – Avg. Popularity: **53.81** (17 titles)| Avg. Rating: **6.11**
5. **English** (en) – Avg. Popularity: **41.76** (7,570 titles)| Avg. Rating: **6.37**

Top Languages by Average Rating (Min. 10 titles)

1. **Swedish** (sv) – Avg. Rating: **7.02** | Avg. Popularity: **21.16** (23 titles)
2. **Japanese** (ja) – Avg. Rating: **6.98** | Avg. Popularity: **40.72** (645 titles)
3. **Russian** (ru) – Avg. Rating: **6.86** | Avg. Popularity: **39.66** (83 titles)
4. **Korean** (ko) – Avg. Rating: **6.78** | Avg. Popularity: **33.51** (170 titles)
5. **Chinese** (zh) – Avg. Rating: **6.74** | Avg. Popularity: **30.02** (129 titles)

[!TIP]

Indian (Hindi) and Turkish content drive massive popularity relative to their small volume, indicating strong viewer draw. Conversely, Japanese and Swedish content are highly rated, indicating strong niche critical acclaim (particularly anime for Japanese).

6. Temporal Trends & Data Anomalies

A. Decade-by-Decade Growth

The dataset highlights an exponential rise in content library size, particularly starting in the 1990s:

- **1900s–1950s:** Under 200 titles total.
- **1980s:** 631 titles (Avg. Rating: 6.55)
- **1990s:** 1,050 titles (Avg. Rating: 6.51)

- **2000s:** 2,093 titles (Avg. Rating: 6.39)
- **2010s:** 3,999 titles (Avg. Rating: 6.43)
- **2020s:** 1,384 titles (Avg. Rating: 6.11 | Avg. Popularity: **107.28**)

[!NOTE]

Average popularity is dramatically higher for 2020s titles because they are newer releases still experiencing high search/streaming traffic.

B. The 2022–2024 Data Anomaly

In the dashboard's "Title by Year" chart, there is a sharp drop in 2022, and the average ratings card shows a depressed score. Analysis of the raw data explains this:

- **Incomplete Data Collection:** The data collection likely cut off in early 2022. There are **208 titles in 2022, 8 in 2023, and only 1 in 2024.**
 - **Zero-Vote Placeholders:** Out of the 217 titles from 2022–2024, **91 titles have exactly 0 votes and 0.0 ratings.** These represent upcoming or unreleased titles at the time of export.
 - **Impact on Averages:**
 - Including placeholders, the average rating for 2022 is an artificially low **3.64**.
 - Excluding the 0-vote placeholders, the average rating for 2022 rises to a normal **6.00**.
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